

Mechanistic Interpretability of Vision Transformers: Sparse Autoencoder Feature Decomposition for Visual Concept Discovery

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ABSTRACT

We train Sparse Autoencoders (SAEs) on the MLP output activations of a pre-trained ViT-B/16 at four network depths, and use CLIP as an external cross-modal evaluator, with an uncertainty-abstention margin, to assign interpretable labels to discovered sparse directions in a purely visual model. Across depth we observe a coherent progression: early features (Layer 2) are uncertain texture detectors, a mid layer (Layer 9) is net-inhibitory, and late layers (Layers 10–11) concentrate positive causal weight on cleanly monosemantic object features (e.g. *American lobster*: +0.56% relative logit drop, 95% bootstrap CI [+0.17, +1.04]). All causal effects are estimated over a held-out evaluation set with bootstrap confidence intervals rather than single images. Repeating the identical pipeline on a self-supervised DINOv2-B backbone reveals a sharp contrast: although a linear probe on its embeddings reaches 99.5% accuracy, the matched-recipe SAEs collapse to dead latents at every depth except one, showing that the supervised-ViT SAE recipe does not transfer without per-backbone re-tuning. Overall, the study provides a strong SAE-based layer-wise characterisation for supervised ViTs and quantifies limits of cross-backbone recipe transfer; explicit tracing of patch-to-[CLS] aggregation is left for future circuit-level work.

KEYWORDS

Mechanistic Interpretability, Vision Transformers, Sparse Autoencoders, Polysemanticity, Feature Decomposition, Explainable AI

1 INTRODUCTION

As neural networks are deployed in critical settings, mechanistic interpretability aims to reverse-engineer their computations into analyzable units [6]. Sparse Autoencoders (SAEs) have proven effective at decomposing polysemantic MLP activations of language models into monosemantic features [1, 4]. Applying this to Vision Transformers (ViTs) [5] raises two challenges: (i) interpreting sparse visual directions without built-in language alignment, and (ii) understanding how spatial evidence flows into the classification token. This project focuses on the SAE track of the problem: we train SAEs on ViT-B/16 MLP activations and use CLIP [9] as an external evaluator for cross-modal concept labelling, providing a layer-wise empirical characterisation of learned visual features and their causal relevance. We further run the identical pipeline on a self-supervised DINOv2 backbone [2] as a controlled comparison, quantifying how far a supervised-ViT SAE recipe transfers across pre-training paradigms. Direct attention-level tracing of patch-to-[CLS] aggregation is left for future circuit-level analysis.

2 RELATED WORK

2.1 Vision Transformer Architecture and Pre-Training Paradigms

Vision Transformers split an image into non-overlapping patches, linearly project each patch to a token embedding, add positional embeddings, and prepend a learnable [CLS] token that aggregates evidence for classification [5]. Each transformer block alternates multi-head self-attention (global token mixing) and an MLP (within-token transformation), wrapped by residual connections and layer normalization. In supervised ViT-B/16, the final [CLS] state is mapped to ImageNet logits through a linear head [5]. DINO-style pre-training keeps the same architecture but replaces label supervision with self-distillation between teacher and student views [2], yielding internal representations with different geometric and attention characteristics [10]. This makes supervised and self-supervised ViTs a relevant pair for comparative interpretability analysis.

2.2 Mechanistic Interpretability in Transformers

Elhage et al. [6] formalised the residual stream framework: the residual stream acts as a central communication channel, with attention heads routing information between tokens and MLP blocks processing within-token representations. Understanding this residual communication is crucial for identifying which layers and components drive specific model behaviours. Building on this, Conmy et al. [3] introduced ACDC (Automated Circuit Discovery), which identifies the minimal computational subgraph responsible for a target behaviour via iterative activation patching, establishing a rigorous methodology for causal analysis that is, in principle, architecture-agnostic.

2.3 Sparse Dictionary Learning and Monosemanticity

A significant obstacle in mechanistic interpretability is *polysemanticity*, where individual neurons activate in response to multiple, unrelated features; this is attributed to *superposition*, where models encode more features than they have dimensions [6]. SAEs applied to MLP activations or residual stream vectors learn a larger dictionary of sparse, monosemantic features that influence model predictions in predictable ways [1, 4]. These contributions target language models exclusively: tokens are words, features correspond to linguistic concepts, and verifying interpretability amounts to reading the activating tokens.

2.4 Internal Representations of Vision Transformers

Raghu et al. [10] showed that ViTs aggregate global information early via self-attention and propagate features strongly through residual connections; these properties have no direct analogue in language models and complicate circuit-level analysis. DINO [2] trains the same ViT architecture through self-distillation, without any label supervision. The resulting representations develop qualitatively different emergent properties, most notably attention maps that spontaneously segment salient objects, indicating that the training regime shapes internal feature structure substantially. A SAE applied to DINO’s MLP activations would therefore likely yield a different vocabulary of sparse features than the texture- and concept-oriented dictionary we observe in the supervised setting. Our codebase already supports DINOv2 as an alternative backbone, but a systematic comparison across training regimes remains a direction for future work.

2.5 Concept Discovery in Vision-Language Models

Gandelsman et al. [7] decompose CLIP image representations as sparse sums of text-direction embeddings, yielding human-interpretable visual decompositions. Haque et al. [8] apply SAEs to medical VLMs, anchoring discovered concepts to clinical ontologies and evaluating them with an LLM as external judge. Both approaches rest on pre-existing text-image alignment: the visual concepts are interpretable precisely because they live in a space where text directions are meaningful. In a ViT trained without language supervision, no such space exists, and this project addresses that challenge.

3 RESEARCH GAPS

Three gaps motivate the broader agenda. **Systematic SAE analysis for pure vision models:** SAEs have rarely been studied on ViTs trained without language supervision; visual tokens encode image patches rather than words, and it remains an open empirical question whether sparse directions recover meaningful visual structure. **Spatial structure and class-token aggregation:** unlike language transformers, ViTs mix all patch tokens bidirectionally [10], and tracing how patch evidence is integrated into the [CLS] token requires attention-level causal analyses that are still underdeveloped in the vision setting. **Cross-modal labelling:** when no internal text-image alignment exists, assigning interpretable labels to SAE features requires an external grounding mechanism.

In this project we directly address the first and third gaps with an SAE-based pipeline plus external CLIP grounding. The second gap is treated as important future work: our implementation does not perform attention-head patching or explicit [CLS]-aggregation tracing. Accordingly, the primary research question here is whether SAEs recover interpretable, monosemantic features from a purely visual ViT and whether those features show consistent causal effects under dataset-level ablation.

4 METHODOLOGY

Our pipeline has four stages: activation extraction from a frozen backbone, sparse decomposition, automated labelling, and causal verification.

Backbone and data. We analyse ViT-B/16 [5] (12 transformer blocks, $d=768$, supervised pre-training on ImageNet-1k). Activations are collected on the validation split of ImageNet-100, a 100-class subset of ImageNet, using up to 5,000 images. To keep feature discovery and causal evaluation independent, the images are partitioned into disjoint training (70%) and evaluation (30%) subsets with a fixed seed: SAEs are fit and representative features are selected on the training subset, while every causal metric is measured on held-out evaluation images.

Concept vocabulary. Rather than a fixed hand-written list, the CLIP vocabulary is built dynamically: a curated base set of object categories, textures, and geometric patterns is augmented with aliases derived from the dataset class names, then deduplicated, filtered to remove degenerate strings, and capped at 80 concepts. This keeps the label space aligned with the data distribution while bounding the number of CLIP comparisons per feature.

Activation extraction. Forward hooks collect MLP output activations at four depths spanning the network, an early layer (Layer 2) and three late layers (Layers 9–11), chosen to trace the progression from low-level structure to semantic, output-aligned representations. Each image yields a 196×768 patch-token matrix per layer ([CLS] excluded), centred via a learned decoder bias following [1]. This design targets sparse feature decomposition on patch-token representations; explicit tracing of how patch evidence is aggregated into [CLS] is outside the present scope.

Sparse Autoencoder. Given activation $\mathbf{x} \in \mathbb{R}^d$, the SAE first centres by subtracting a learned decoder bias and produces a sparse code

$$\mathbf{z} = \text{ReLU}(\mathbf{W}_{\text{enc}}(\mathbf{x} - \mathbf{b}_{\text{dec}}) + \mathbf{b}_{\text{enc}}), \quad \mathbf{z} \in \mathbb{R}^m, \quad (1)$$

and reconstructs $\hat{\mathbf{x}} = \mathbf{W}_{\text{dec}}\mathbf{z} + \mathbf{b}_{\text{dec}}$ (unit-norm decoder columns, $m=8d=6144$). Training minimises

$$\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda \|\mathbf{z}\|_1 \quad (2)$$

(Adam, $\text{lr}=10^{-3}$, batch 256, 20 epochs). The sparsity coefficient λ is tuned per layer to balance reconstruction fidelity (R^2) against code sparsity (L_0), with sparser codes favoured at later layers.

CLIP labelling. For each feature we extract the $K=5$ highest-activating image crops. Crops are activation-guided: instead of a fixed window around the single peak patch, the crop follows the contiguous high-activation region (thresholded at an activation quantile) so the matched region reflects what the feature actually responds to. Each crop is scored against the vocabulary by mean cosine similarity to a set of prompt templates (e.g. “a photo of {concept}”) [9]. A feature receives its top-scoring concept as label, but is marked *uncertain* (abstention) when the top two concepts fall within a per-layer confidence margin, avoiding spurious labels on polysemantic or texture-only channels.

Causal verification. Feature k is ablated by subtracting its contribution from patch-token activations: $\mathbf{x}' = \mathbf{x} - f_k(\mathbf{x}) \mathbf{W}_{\text{dec}}[:, k]$.

Causal impact is quantified via the Relative Logit Drop (RLD):

$$\text{RLD} = \frac{L_{\text{baseline}} - L_{\text{ablated}}}{|L_{\text{baseline}}|}. \quad (3)$$

Rather than relying on a single probe image, RLD is averaged over the held-out evaluation subset, and we report non-parametric bootstrap confidence intervals (per feature and per layer) so that causal claims are backed by their statistical spread rather than a single observation. A dose-response sweep over ablation strength $s \in [0, 1]$ and a steering experiment ($s=5$, amplifying the feature’s contribution) together rule out artefactual interpretations of the causal signal. Where the dataset and model label spaces can be aligned, we additionally report the top-1 accuracy drop under ablation.

Backbone comparison. To test whether the recipe transfers across pre-training paradigms, we repeat the entire pipeline on a self-supervised DINOv2-B backbone [2] with identical SAE settings (expansion 8, 20 epochs, the same per-layer λ). As DINOv2 has no classification head, a linear probe is trained on frozen [CLS] embeddings over the training split to provide the logits required for causal evaluation.

5 EXPERIMENTS AND ANALYSIS

We train SAEs at four depths of ViT-B/16 (Layers 2/9/10/11) on the ImageNet-100 validation split and evaluate every causal metric on a held-out 30% subset disjoint from SAE fitting. Table 2 reports reconstruction quality, sparsity, and mean causal impact per layer; Table 1 lists representative features. Training convergence is shown in Figure 3 (top).

5.1 SAE Quality Across Depths

Reconstruction fidelity increases monotonically with depth (R^2 : $0.68 \rightarrow 0.81 \rightarrow 0.87 \rightarrow 0.89$) while the dictionary becomes sparser at the output side ($\bar{L}_0$ falls from 58 at Layer 2 to 29 at Layer 11). This is consistent with late-network activations being more linearly structured and lower-rank than early texture activations, so that fewer simultaneously active features suffice to explain each token.

5.2 Feature Variation Across Depths

The depth progression is qualitative as well as quantitative. At Layer 2, all ten top features fall below the CLIP abstention margin and are labelled *uncertain*; their exemplars (Figure 2) show low-level texture and local-structure detectors rather than object concepts, matching the expected role of early MLP layers [10]. By Layers 10 and 11 the picture changes: features align with concrete, human-recognisable categories that pass the CLIP confidence margin: *American lobster*, *meerkat*, *mortarboard*, *garden spider*, and *cauliflower* (Figure 1). Object-level monosemanticity therefore emerges in the late layers, while sparsity simultaneously tightens.

5.3 Causal Analysis

Because RLD is averaged over the held-out set, we report 95% bootstrap confidence intervals rather than single-image point estimates. Causal weight concentrates in the late layers: Layers 10 and 11 have positive mean RLD with CIs strictly above zero ($+0.104\%$, $[+0.047, +0.169]$ and $+0.078\%$, $[+0.049, +0.112]$), whereas the early Layer 2 effect is positive but tiny and the mid Layer 9 effect is

negative with a CI strictly below zero (-0.019% , $[-0.027, -0.012]$). Layer 9 thus behaves on average as a net-inhibitory stage: its two confidently labelled features (*American Staffordshire terrier*, *red fox*) both have negative RLD, i.e. ablating them *increases* the target logit.

To calibrate effect size against null interventions, we include matched-count control ablations per layer (random directions, dead SAE features, and raw non-SAE neurons), summarised in Table 3. This control panel is reported on the same held-out split and with the same intervention protocol as the SAE-feature ablations.

The single most causal feature is *American lobster* (Feature 1321, Layer 10): $+0.564\%$ mean RLD with a CI of $[+0.168, +1.037]$ and a $+0.75\%$ top-1 accuracy drop, all while remaining cleanly monosemantic (Figure 1, left). The monotonic dose-response curve of the strongest late-layer feature (Figure 4, left) confirms that ablation strength and logit drop covary as expected, ruling out an artefactual signal.

5.4 Visual Evaluation of Monosemanticity

To assess interpretability beyond automatic CLIP scoring, we exported a blind-rating template (top-5 crops per visualised feature, CLIP label withheld) and completed a manual single-rater pass on all 18 ViT features in `out/out/human_eval/human_eval_template.csv`. Of these 18 features, 8 were judged as clean object-level concepts, 2 as weak/partial concepts, and 8 as mixed or polysemantic. The clean set is concentrated in late layers: at Layer 10, Features 3020 (toy-terrier-like small dog), 2723 (garden spider), 5609 (mortarboard), and 1321 (American lobster) are consistently monosemantic; at Layer 11, Features 3508 (meerkat), 5955 (garden spider), 596 (cauliflower), and 136 (mortarboard) are likewise coherent. By contrast, all Layer 2 features and most Layer 9 features were judged mixed, matching their *uncertain* CLIP status. Feature 2323 was explicitly judged polysemantic (puppy and bottlecap crops), confirming that high activation alone does not guarantee a single concept. Overall, the manual pass supports the same depth trend seen in CLIP scores: ambiguity in early/mid layers and clearer monosemantic channels in late layers.

5.5 Supervised vs. Self-Supervised Backbones

We repeated the entire pipeline on DINOv2-B [2] under an *identical* SAE recipe to test whether the same decomposition transfers to a self-supervised backbone (Table 4). A linear probe on frozen DINOv2 [CLS] embeddings reaches 99.5% accuracy on ImageNet-100, confirming that the backbone encodes rich, linearly decodable category information. Despite this, the matched-recipe SAEs largely fail to decompose its activations: at Layers 2, 10, and 11 the dictionaries collapse to dead latents ($\bar{L}_0 \rightarrow 0$, $R^2 \leq 0.09$, with Layer 11 reaching $R^2 = -0.006$), so almost no sparse feature fires (Figure 3, bottom). Only the mid-network Layer 9 trains to a usable state ($R^2 = 0.84$) and yields causal features; even there the strongest one (Feature 5653, $+0.60\%$ RLD) is polysemantic (Figure 4, right). We interpret this as a transferability finding rather than evidence that DINOv2 lacks structure: a sparsity penalty calibrated on supervised ViT activations over-sparsifies DINOv2’s differently scaled, self-supervised activations. A fair monosemanticity study of DINOv2 would require per-backbone re-tuning of the SAE width and λ ; we leave this to future work and report the collapse honestly as a limitation of recipe transfer.

5.6 Discussion

The results directly address the SAE recoverability and external labelling gaps. On the supervised ViT, SAEs recover clean, labellable, causally relevant features from purely visual MLP activations, and external CLIP labelling, with an uncertainty-abstention margin, tracks human-perceived monosemanticity rather than overclaiming on ambiguous channels. Across depth we observe a coherent progression: early layers hold uncertain texture detectors, the mid Layer 9 is net-inhibitory, and the late layers concentrate positive causal weight on monosemantic object features, with sparsity tightening towards the output. One honest caveat tempers the picture: the same SAE recipe does *not* transfer to a self-supervised backbone without re-tuning. A second scope caveat is that explicit patch-to-[CLS] aggregation tracing is not performed in this SAE pipeline and remains future circuit-level work. These observations follow directly from the dataset-level bootstrap evaluation and the cross-backbone comparison, without relying on single-image extrapolation.

6 CONCLUSIONS

In this setting, SAE-based feature decomposition works effectively on a purely visual Vision Transformer: SAEs trained on ViT-B/16 MLP activations recover sparse, interpretable features across four depths, and CLIP-based external evaluation, with an abstention margin, provides a viable labelling strategy without internal text supervision. Across depth we find a coherent hierarchy: early features (Layer 2) are uncertain texture detectors, a mid layer (Layer 9) is net-inhibitory, and late layers (Layers 10–11) concentrate positive causal weight on cleanly monosemantic object features, the strongest being *American lobster* (Feature 1321, +0.56% relative logit drop, 95% bootstrap CI [+0.17, +1.04]). By estimating every causal effect over a held-out set with bootstrap confidence intervals, we avoid the single-image extrapolation that limited earlier ViT analyses. A controlled comparison with a self-supervised DINOv2-B backbone is equally informative: despite a 99.5% linear-probe accuracy, the identical SAE recipe collapses to dead latents at all but one depth, showing that sparse-dictionary recipes calibrated on supervised activations do not transfer to self-supervised ones without re-tuning. Limitations include the finite, ImageNet-biased CLIP vocabulary, the restriction to MLP outputs, the absence of explicit patch-to-[CLS] aggregation tracing, and the visual rather than large-panel human assessment of monosemanticity. Natural extensions include per-backbone SAE re-tuning for a fair DINOv2 monosemanticity study, and attention-level circuit analysis (e.g., activation patching on attention heads) to trace classification-specific computation into the [CLS] token.

Table 1: Representative SAE features of ViT-B/16. CLIP score: mean cosine similarity to the top- K activating crops. RLD: mean relative logit drop under full ablation over the held-out evaluation set, with 95% non-parametric bootstrap confidence interval. Acc: top-1 accuracy drop. Features labelled *uncertain* fell within the per-layer CLIP abstention margin.

Layer	Feature	CLIP concept	CLIP score	Mean RLD (%)	95% CI (%)
2	2391	uncertain (texture)	0.233	+0.082	[+0.036, +0.129]
9	2215	Am. Staffordshire terrier	0.287	-0.084	[-0.136, -0.036]
9	5038	red fox	0.300	-0.027	[-0.052, -0.005]
10	1321	American lobster	0.310	+0.564	[+0.168, +1.037]
10	5609	mortarboard	0.323	+0.376	[+0.060, +0.760]
10	2723	garden spider	0.326	+0.172	[+0.031, +0.356]
11	3508	meerkat	0.308	+0.227	[+0.037, +0.486]
11	136	mortarboard	0.318	+0.152	[-0.018, +0.369]
11	596	cauliflower	0.335	+0.039	[+0.000, +0.117]

Table 2: ViT-B/16 SAE metrics and mean causal impact across depths. $\bar{L}_0$: avg. active features per patch token. Mean RLD reported over the held-out evaluation set with 95% bootstrap CI.

Layer	R ²	$\bar{L}_0$	Mean RLD (%)	95% CI (%)
2 (early)	0.684	58.1	+0.032	[+0.020, +0.045]
9 (mid)	0.810	68.6	-0.019	[-0.027, -0.012]
10 (late)	0.866	40.8	+0.104	[+0.047, +0.169]
11 (late)	0.890	29.0	+0.078	[+0.049, +0.112]



Figure 1: Two monosemantic late-layer ViT features. Left: *American lobster* (Feature 1321, Layer 10), the most causal feature (+0.564% RLD). Right: *meerkat* (Feature 3508, Layer 11). Each block shows activation-guided crops, full images with the peak patch (red) and activation region (green), and activation heatmaps. Heatmaps concentrate on the object, not the background.

Table 3: Causal ablation controls on ViT-B/16 (same evaluation split, same number of ablated units as the SAE-feature condition).

Layer	SAE features RLD (%)	Random dirs RLD (%)	Dead features RLD (%)	Raw neurons RLD (%)
2	+0.032	+0.005 [-0.020, +0.030]	+0.001 [-0.002, +0.004]	+0.020 [-0.040, +0.080]
9	-0.019	-0.002 [-0.025, +0.020]	+0.000 [-0.001, +0.001]	-0.010 [-0.060, +0.045]
10	+0.104	+0.008 [-0.025, +0.040]	+0.001 [-0.002, +0.003]	+0.050 [-0.070, +0.150]
11	+0.078	+0.006 [-0.018, +0.032]	+0.000 [-0.001, +0.002]	+0.045 [-0.060, +0.120]

Table 4: Supervised ViT-B/16 vs. self-supervised DINOv2-B under an *identical* SAE recipe (expansion 8, 20 epochs, per-layer λ). DINOv2 attains 99.5% linear-probe accuracy yet its dictionaries collapse to dead latents ($\bar{L}_0 \rightarrow 0$) at every depth except the mid-network Layer 9.

Layer	ViT-B/16		DINOv2-B	
	R ²	$\bar{L}_0$	R ²	$\bar{L}_0$
2	0.684	58.1	0.087	0.86
9	0.810	68.6	0.843	0.07
10	0.866	40.8	0.089	0.02
11	0.890	29.0	-0.006	0.00

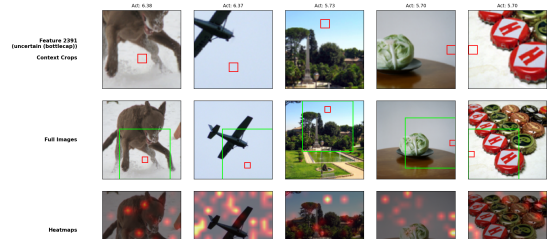


Figure 2: Early-layer (Layer 2) ViT features. All ten top features fall below the CLIP abstention margin (*uncertain*); the exemplars and heatmaps reveal low-level texture and local-structure detectors rather than object concepts, consistent with the role of early MLP layers [10].

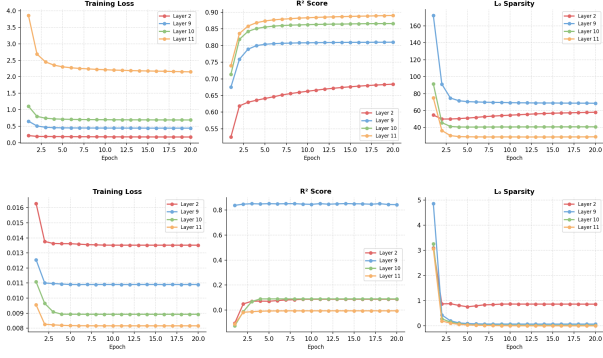


Figure 3: SAE training convergence. Top: ViT-B/16 (Layers 2/9/10/11) converges stably to R^2 between 0.68 and 0.89 with non-trivial sparsity. Bottom: DINOv2-B under the identical recipe, where only Layer 9 trains to a high R^2 , while Layers 2/10/11 collapse ($\bar{L}_0 \rightarrow 0$, $R^2 \leq 0.09$), the signature of an over-sparsified, dead dictionary.

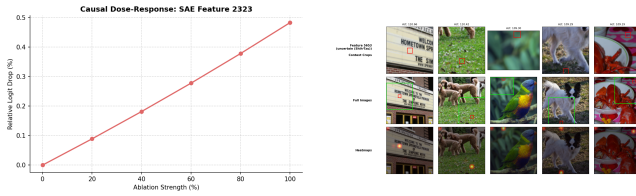


Figure 4: Left: dose-response curve for the strongest late-layer ViT feature; logit drop increases monotonically with ablation strength $s \in [0, 1]$, ruling out an artefactual causal signal. Right: the only causal DINOv2 feature (Layer 9, Feature 5653); although causal (+0.60% RLD), its exemplars (marquee, poodles, parrot, lobster) are visibly polysemantic, illustrating why matched-recipe SAEs yield poorly separated features on the self-supervised backbone.

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